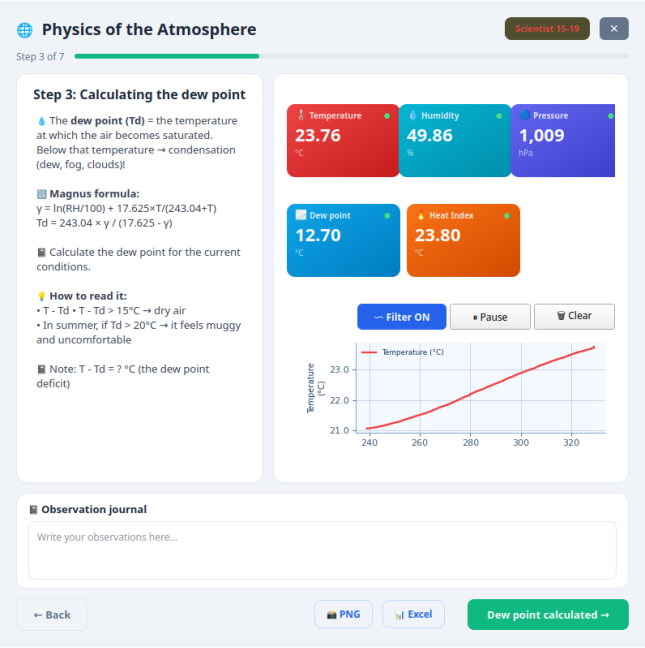
Physics of the Atmosphere

A single pressure sensor becomes a weather station: altitude, dew point and the scale height of the air

Level: Scientist — ages 15–19 (Romanian school years IX–XII) • **Time in class:** 50 min, two 50-minute lessons or three 50-minute lessons • **Sensors:** BME280 (temperature, humidity, pressure)



The activity screen. The values in the picture are simulated; your own will differ.

Before you start

Who it is for

Scientist — ages 15–19 (Romanian school years IX–XII). Everything that uses logarithms — the $\ln P$ column, the gradient, the Magnus formula — is for **year X and up**, once logarithms have been taught. The short route, with no derivation and no logarithms, also works in years VIII and IX: there it remains an experimental check that pressure falls by about 0.12 hPa per metre.

Assumed knowledge

Natural logarithm (year X), reading a gradient off a graph, the equation of a line $y = c + mx$. For the full route, in addition: the ideal gas equation of state. For the short route: only reading a graph.

Time in class

Short: 50 min. Full: two 50-minute lessons (1 h 40 min). Project: three 50-minute lessons (2 h 30 min).

Teacher preparation

15 min — choosing and walking the staircase, measuring the height of one storey, noting the school's true altitude and today's QNH value, printing the sheets with the $\ln P$ column and the axes already prepared.

Sensor settling time

2–3 minutes after each move, until the pressure settles. This is the most common reason for bad data.

Sensors and boxes

BME280. One box per group; at least two groups, so you can compare.

Three routes

Short	50 min, no Heat Index and no full derivation: prediction, pressure at two levels, dew point, conclusion.
Full	Two 50-minute lessons: full derivation, Heat Index and QNH in the first; the stairwell traverse, the $\ln P$ graph and its gradient in the second.
Project	Three 50-minute lessons: the two above, plus the written-analysis lesson — the scale height H, the comparison with 8.4 km, the molar mass of air.

1. Where this fits in the curriculum

This is the most quantitative activity in the collection. Students feed three sensor readings into four physical models, then test one of those models by climbing a staircase. It suits:

- **Physics** (*Fizică*) — at **year X**, where the syllabus covers molecular physics and thermodynamics, the activity sits exactly on top of the thermal equation of state for an ideal gas and on the adiabatic transformation; fluid pressure and atmospheric pressure are revision from lower-secondary school (gimnaziu). At **year IX** it can be used as an application of mechanical equilibrium — fluid statics remains lower-secondary material, not an entry in the year IX syllabus. At **year XII**, the exponential law obtained here is exactly the form of the radioactive-decay law, and students who have once linearised a curve with a logarithm will recognise it immediately.
- **Mathematics** (*Matematică*) — at **year X**, the exponential function and the logarithmic function, the properties of the logarithm, exponential and logarithmic equations: here they gain a measurable purpose. The gradient of a line and the linear function come from **year IX**. At **years XI–XII**, the derivative and the definite integral allow the full derivation to be worked through, from $\mathrm{d}P/\mathrm{d}z = -\rho g$ to the barometric formula.
- **Geography** (*Geografie*) — at **year IX**, general physical geography covers the Earth's atmosphere: its structure, atmospheric pressure, air temperature and humidity, condensation, the synoptic chart and forecasting. The activity produces, with classroom instruments, the very quantities shown on the synoptic chart.
- **Computer science and ICT** (*Informatică și TIC*) — **years IX–XII**, processing data in a spreadsheet: the computed $\ln P$ column, the XY chart, the regression line and its gradient. The file the app exports opens directly in a spreadsheet program.
- **Cross-curricular** — the distinction between *accuracy* and *precision*, the domain of validity of an empirical model, the propagation of uncertainty, and the honesty of writing down what the instrument showed, not what it should have shown.

The central intellectual move is this: **an exponential law becomes a straight line if you take the logarithm**, and the gradient of that line is a physical quantity you can name. A student who understands this once will reuse it for radioactive decay, capacitor discharge, absorbance and population growth.

2. Time plan

The in-app wizard is written as a single run of about 50 minutes in one fixed location. The genuinely memorable version adds a vertical traverse of the building, which needs a second lesson, and a written analysis, which needs a third. Choose the row that matches your timetable.

Route	Time	What happens
Short	**50 min**	Written predictions (7), theory in brief, no derivation (6), the baseline measurement and altitude computed by hand (7), the dew point (10), the pressure comparison between two levels (12), conclusion and comparison with the predictions (8). **No Heat Index and no full derivation.** This is the route that suits an ordinary lesson.
Full, lesson 1	**50 min**	Written predictions (7), the full hydrostatic derivation on the board (12), the baseline measurement and altitude (7), the dew point (8), Heat Index (6), QNH and the calibration check (6), closing (4).
Full, lesson 2	**50 min**	Measuring the height of one storey with a tape measure, adopted by the whole class (8); the stairwell traverse, with a stop on every landing (25); the $\ln P$ against height graph, its gradient and a discussion of what limited the result (17). Data is exported at the end.
Project, lesson 3	50 min	Written analysis: the regression line, the scale height $H = -1/m$, the comparison with the theoretical value of 8.4 km expressed as a percentage, one named source of systematic error and, as an extension, the mean molar mass of air.

The full route means **two 50-minute lessons**, and the project route adds the third, so **three 50-minute lessons** in total — two and a half hours of contact time, ideally on different days. Your own preparation takes about a quarter of an hour the day before (see Materials).

Pacing warning — sensor settling

The BME280 needs **30–60 seconds to settle** after it is moved: the pressure reading is smoothed (by the sensor's filter and by the app's averaging), so it reaches the new value gradually, and a warm sensor carried into a cold stairwell reads a *temperature* that keeps drifting for a minute or two. Allow **2–3 minutes per landing** in practice. Make this a stated habit — "wait, watch the number until it stops moving, only then record" — because groups who record instantly produce noise, not data. With ninety seconds of walking and two minutes of waiting per landing, a four-storey traverse fits into the plan's 25 minutes.

3. Materials

Item	Per group	Notes
MagicBox with BME280	1	The only sensor required. Its pressure resolution is far finer than its absolute accuracy, and it is exactly that distinction the activity exploits.
Laptop with the app	1	For the traverse the laptop travels with the box, so a charged battery matters. A USB cable is more reliable than WiFi in a concrete stairwell.
Stairwell or building with several floors	1	Shared between groups. Four floors or more is ideal; two floors still works, but the signal is small. A lift shaft, a hill or a tall block of flats are all acceptable substitutes.
Tape measure or long ruler	1	To measure the rise of a flight of stairs or the floor-to-floor height. A 5 m builder's tape is ideal.
Today's QNH value, printed	1	Sea-level pressure for your town, taken from the national weather service and written on the board. This is the only external reference the class compares against.
The school's true altitude	1	Look it up in advance on a topographic map or a mapping service, and keep it to yourself until after the students have calculated their own.
Clipboard, pencil and pocket calculator	1 / student	Recording is done standing on a staircase; a loose sheet is a lost sheet. The calculator must have ln and x^y keys.

Preparation the day before: walk the route yourself and decide where each group stops, so that thirty students are not standing on the same landing. Measure the floor-to-floor height once, exactly, so you have a reference value against which to check the groups' measurements. Look up two numbers and write them on a card you keep in your pocket: the **school's true altitude above sea level** and **today's QNH** from the national weather service. Finally, switch a box on and leave it on your desk for ten minutes to confirm that the pressure reading is stable to within a few hundredths of a hPa; a sensor that jitters by more than about 0.1 hPa at rest will not be able to distinguish a single storey.

4. Scientific background for the teacher

Why pressure falls exponentially with height

The atmosphere is not held up by anything. Every layer of air is squeezed by the weight of all the air above it, and the pressure at any height is simply that weight per unit area. Take a thin horizontal slab of air, of thickness $\mathrm{d}z$ and density ρ. Balancing the weight of the slab against the pressure difference across it gives the **hydrostatic equation**:

$$\frac{\mathrm{d}P}{\mathrm{d}z} = -\rho g.$$

Air behaves very nearly like an ideal gas, so with molar mass M and gas constant R,

$$P = \rho \frac{R}{M} T \qquad \Longrightarrow \qquad \rho = \frac{PM}{RT}.$$

Substituting and separating the variables:

$$\frac{\mathrm{d}P}{P} = -\frac{Mg}{RT}\,\mathrm{d}z \equiv -\frac{\mathrm{d}z}{H}, \qquad H \equiv \frac{RT}{Mg}.$$

If the temperature were constant with height, integrating gives the **barometric formula** in its simplest form:

$$\boxed{P(z) = P_0\,\mathrm{e}^{-z/H}} \qquad \text{or, equivalently,} \qquad \ln P = \ln P_0 - \frac{z}{H}.$$

The quantity H is the **scale height**: the vertical distance over which pressure falls by a factor of e. With $R = 8.314\ \mathrm{J\,mol^{-1}K^{-1}}$, $T = 288\ \mathrm{K}$, $M = 0.0290\ \mathrm{kg\,mol^{-1}}$ and $g = 9.81\ \mathrm{m\,s^{-2}}$,

$$H = \frac{8.314 \times 288}{0.0290 \times 9.81} \approx 8.4 \times 10^{3}\ \mathrm{m}.$$

This is the single most important number in the activity. It says that the atmosphere, for all its apparent vastness, is effectively only about eight kilometres thick — thinner, relative to the Earth, than the skin of an apple.

4.1 The linearisation, and why it matters

The second form of the boxed result is the whole point of the analysis. An exponential curve is hard to check by eye: almost any smooth decreasing curve looks plausible. But

$$\ln P = \ln P_0 - \frac{1}{H}\,z$$

is the equation of a straight line, $y = c + mx$, with

$$y = \ln P, \qquad x = z, \qquad m = -\frac{1}{H}.$$

So if students plot the natural logarithm of the pressure against the height above their starting point and the points fall on a straight line, they have confirmed the exponential law. The gradient then hands them the scale height directly:

$$H = -\frac{1}{m}.$$

From a fifteen-metre climb inside the school building, students measure a property of the entire atmosphere. It is worth saying that out loud in class.

4.2 How big is the signal in a stairwell

Over a stairwell $z \ll H$, so the exponential can be expanded to first order:

$$P(z) = P_0\left(1 - \frac{z}{H} + \dots\right) \qquad \Longrightarrow \qquad \frac{\Delta P}{\Delta z} \approx -\frac{P_0}{H} = -\frac{1013\ \mathrm{hPa}}{8400\ \mathrm{m}} \approx -0.12\ \mathrm{hPa\,m^{-1}}.$$

This is exactly the "rule of thumb" the app uses at the QNH step. A storey of about 3 m therefore changes the pressure by roughly 0.36 hPa, and a four-storey climb by about 1.5 hPa. The BME280 resolves changes of order 0.01 hPa, so the effect is comfortably measurable — *provided* students compare readings from the *same* sensor, taken within a few minutes, and never compare one group's absolute reading with another's.

Relative precision versus absolute accuracy

The BME280's absolute pressure accuracy is of order ±1 hPa — that is, roughly ±8 m, nearly three times a storey — while its ability to detect a *change* is around 0.01 hPa. The activity therefore illustrates the accuracy–precision distinction perfectly: a measurement can be simultaneously too crude to give the correct altitude and sharp enough to extract the scale height of the atmosphere.

4.3 Where the app's altitude formula comes from

The real troposphere is not isothermal: temperature falls with height at an average rate $L \approx 6.5\,^{\circ}\mathrm{C}$ per kilometre (the *environmental lapse rate*; the *dry adiabatic lapse rate* quoted in the app's theory step, 9.8 °C per kilometre, applies to an unsaturated parcel of air that is rising, not to the ambient profile). Writing $T(z) = T_0 - Lz$ and integrating again,

$$\int_{P_0}^{P} \frac{\mathrm{d}P}{P} = -\frac{Mg}{R}\int_{0}^{z} \frac{\mathrm{d}z}{T_0 - Lz} \qquad \Longrightarrow \qquad P = P_0\left(1 - \frac{Lz}{T_0}\right)^{Mg/(RL)}.$$

Rearranging for the height gives the formula the students use:

$$z = \frac{T_0}{L}\left[1 - \left(\frac{P}{P_0}\right)^{RL/(Mg)}\right] = 44330\left[1 - \left(\frac{P}{1013.25}\right)^{0.1903}\right]\ \mathrm{m}.$$

Both constants are worth unpacking with a strong class, because they are not arbitrary:

$$\frac{T_0}{L} = \frac{288.15\ \mathrm{K}}{0.0065\ \mathrm{K\,m^{-1}}} = 44330\ \mathrm{m}, \qquad \frac{RL}{Mg} = \frac{8.314 \times 0.0065}{0.02896 \times 9.807} = \frac{1}{5.256} = 0.1903.$$

So 44330 is the height at which the standard atmosphere's temperature profile would reach absolute zero, and 0.1903 is a pure combination of R, L, M and g. Nothing has been fitted to data.

4.4 The reference-pressure trap, and QNH

The formula above assumes $P_0 = 1013.25$ hPa, the *standard* sea-level pressure. On a real day, the actual sea-level pressure might be 995 or 1030 hPa. Since $\mathrm{d}z/\mathrm{d}P \approx -8.3$ m per hPa near the ground, a 10 hPa error in the assumed reference produces an error of roughly 83 m in the computed altitude. Students who get "our town is at −40 m" have not made an arithmetic mistake; they have discovered why aircraft altimeters must be reset to the local pressure before every landing. With 1013.25 hPa in the formula, the absolute altitude can be off by ±100–250 m; with the day's QNH instead — the value you can also enter in **Settings**, in the **Sea-level pressure (QNH)** field — it comes out right to within a few metres.

The inverse operation — taking a station's measured pressure at a known height h and reducing it to sea level — gives **QNH**, the quantity every weather map actually shows:

$$\mathrm{QNH} = P\left(1 + \frac{0.0065\,h}{T + 0.0065\,h + 273.15}\right)^{5.257} \qquad \text{or, near the ground,} \qquad \mathrm{QNH} \approx P + 0.12\,h.$$

Reducing to sea level is what makes pressures from stations at different altitudes comparable, and it is why a weather map shows smooth isobars rather than a picture of the terrain. If a group's QNH matches the national forecast to within two or three hPa, their sensor is well calibrated; this is the only external check available in the classroom and it should be done every time.

4.5 The dew point and the Magnus formula

Saturation vapour pressure rises steeply and non-linearly with temperature. The Magnus approximation to the Clausius–Clapeyron relation is

$$e_s(T) = 6.112\,\exp\!\left(\frac{aT}{b+T}\right)\ \text{hPa}, \qquad a = 17.625, \quad b = 243.04\,^{\circ}\text{C}.$$

Relative humidity is the ratio of the actual vapour pressure to the saturation value, $\text{RH} = 100\,e/e_s(T)$. The **dew point** T_d is the temperature at which the vapour already present in the air would be saturated, that is, $e_s(T_d) = e$. Taking logarithms of both sides and solving gives exactly the expression on screen:

$$\gamma = \ln\frac{\text{RH}}{100} + \frac{aT}{b+T}, \qquad T_d = \frac{b\,\gamma}{a-\gamma}.$$

Worked example for an ordinary classroom, $T = 22.0\,^{\circ}\text{C}$ and $\text{RH} = 45\%$: $\gamma = \ln 0.45 + (17.625 \times 22)/(243.04 + 22) = -0.799 + 1.463 = 0.664$, hence $T_d = 243.04 \times 0.664/(17.625 - 0.664) = 9.5\,^{\circ}\text{C}$. The **dew point deficit** $T - T_d = 12.5\,^{\circ}\text{C}$ indicates comfortably dry air. A deficit below about $5\,^{\circ}\text{C}$ makes fog likely; a dew point above $20\,^{\circ}\text{C}$ in summer is what people call "muggy". The physically important point is that *the dew point, unlike relative humidity, does not change if you merely heat or cool the air*. It tracks the amount of water actually present, which is why meteorologists prefer it. The app uses the same constants everywhere, $a = 17.625$ and $b = 243.04\,^{\circ}\text{C}$, so the hand calculation and the card should agree to a tenth of a degree.

4.6 The Heat Index, and its honestly stated limits

The Heat Index card combines temperature and humidity into an apparent temperature, using the Rothfusz regression to Steadman's apparent-temperature tables. In degrees Celsius, the published coefficients are

$$\begin{aligned}\text{HI} = {} & -8.78469 + 1.61139\,T + 2.33855\,\text{RH} - 0.14612\,T\,\text{RH} - 0.012308\,T^2 \\ & - 0.016425\,\text{RH}^2 + 0.0022117\,T^2\text{RH} + 0.00072546\,T\,\text{RH}^2 - 0.0000035820\,T^2\text{RH}^2.\end{aligned}$$

On screen the formula appears rounded to three significant figures (H is the relative humidity, in %), so the two terms with H^2 read there $-0.0164\,\text{H}^2$ and $-0.000003582\,T^2\text{H}^2$; its range of validity is written beside it: $T \geq 27\,^{\circ}\text{C}$ and $\text{H} \geq 40\%$. This is a *regression*, not a law: it was fitted to tabulated values above roughly $27\,^{\circ}\text{C}$ and 40% relative humidity, and it has no meaning outside that range. In an ordinarily heated classroom, at 20–24 °C, the model simply does not apply. The app knows this and, below the threshold, the card shows the air temperature itself — see the technical note beside that step in the lesson flow. Do not let students quote the card as a result below the threshold: have them compute the Heat Index by hand from the coefficients above, see what value comes out, and treat "is the model being used inside its valid range?" as a discussion question. A model with an honestly stated domain of validity is better science than a formula applied everywhere.

5. Lesson flow

The in-app wizard has **ten steps**: theory; a **written prediction**; baseline data; dew point; heat index; QNH; a second prediction, before the monitoring; continuous monitoring; the **How sure are we? Scatter, and a control reading** step; and the conclusion. The app keeps the predictions in the pupil's journal and quotes them back in the conclusion. The live screen shows the **Temperature**, **Humidity** and **Pressure** cards (the pressure to two decimal places at Scientist level), plus the computed **Dew point** and **Heat Index**, above a chart of temperature and pressure since the start of the step. At the QNH step a line under the cards also gives the QNH the app computes from the school altitude entered in **Settings** ("With the school altitude from Settings (h = … m): QNH ≈ … hPa"), so pupils can check their hand calculation. Below is

what the teacher does around those steps. Each step is marked with the route it belongs to. **The short route skips steps 5 and 6** — Heat Index and QNH — and **the full derivation is set aside inside step 2**, where the result is stated without being derived.

1. **Ask for the written predictions first (7 min — both routes).** Before any box is switched on, every student writes three answers: *does air pressure increase or decrease as you climb, and by how much per storey? Will the change be big enough for our sensor to see? Roughly what is the dew point of the air in this room?* For the second question, insist on a number with a unit. Most classes guess wildly — some by a factor of a thousand — and that spread is the hook for the lesson. Use the extra minutes to hear two or three different justifications; the prediction also goes into the app's prediction box.

2. **Theory (6 min on the short route; 12 min on the full route).** On the full route, work through the hydrostatic derivation on the board as far as $P = P_0 \mathrm{e}^{-z/H}$ and the value of H; do not present the 44330 formula as magic, show where the two constants come from. **On the short route, skip the derivation:** state the result — pressure falls exponentially, roughly 0.12 hPa per metre — and move on to measuring. Then work through the wizard's first screen together.

Technical note — the sensor and the buttons, right before the first reading

The sensor. BME280, which gives temperature, relative humidity and barometric pressure from a single chip. Nothing extra is plugged in. Pressure is the quantity the whole activity depends on: its resolution (about 0.01 hPa) is far better than its absolute accuracy (±1 hPa), which is why you compare *differences*, never absolute values, between boxes. At Scientist level pressure is shown to two decimal places, precisely so that small differences are visible; the pressure chart spans at least 1 hPa vertically, so that noise does not look like a signal.
How long to wait. 30–60 seconds after each move, and the same again after a change of temperature. In the stairwell, allow 2–3 minutes per landing. This is the single thing that most often ruins this activity's data.
How to start. From the **Activities** page, with the **Start** button on the activity card.
The buttons you will use. The activity moves forward with the green button at the bottom right, and **its label changes at every step**, because it echoes what you have just finished: **I understand the theory!**, **My prediction is written**, **Baseline data noted**, **Dew point calculated**, **Heat Index calculated**, **QNH calculated**, **My prediction is written** again, **Changes monitored**, **Understood**, and at the last step **Finish**. At the baseline and dew-point steps the **Capture** button keeps the reading, which the conclusion quotes. The **Back** button, on the left, goes back one step. Above the chart sit **Smooth: on / Smooth: off**, **Pause** — which freezes the chart, useful when you want to show it to the class — and **Clear**.
Exporting the data. Two buttons appear on the measurement steps: **PNG** saves the charts as an image, and **Excel** saves the step's *raw* (unfiltered) readings to an `.xlsx` file, with an `elapsed_s` column (seconds since the start of the step) — this is the file in which you will build the $\ln P$ column. The pupil's predictions and journal notes are saved automatically, to disk (in the `logs` folder, a `…_journal.txt` file) and as a **Journal** sheet in the activity's Excel file. The box's complete recording is taken from the **Data & Export** page, which is visible at Scientist level too; exports default to `.xlsx`, and a CSV saved while the app is in Romanian, French or German uses ";" and a decimal comma. Logs on disk are no longer deleted automatically. Save using the pattern `class_date_activity` — for example `10C_2026-09-14_atmosfera`. Note on the sheet the place (building and starting floor), the time, the measured height of one storey and the forecast QNH value, and check that the file really was saved before closing the application.

3. **Baseline measurement and altitude (7 min — both routes).** Boxes sit on desks, undisturbed, for at least a minute. Students record T, P and RH, then compute their altitude

by hand with the barometric formula — once with 1013.25 hPa and once with the day's QNH — and compare it with the school's true altitude, which you reveal only now. Expect a mismatch of tens of metres and treat that mismatch as the lesson's first real result, not an error. Read the technical note below *before* discussing the figure with the class.

Technical note — right at the step where students compute altitude

The computed altitude depends on the assumed reference pressure, not only on where you are. The app has no altitude card: students obtain it themselves, from the barometric formula shown at the measurement step. The barometric formula used here assumes $P_0 = 1013.25$ hPa, the *standard* sea-level pressure. On a real day, the actual sea-level pressure might be 995 or 1030 hPa. Because near the ground a 1 hPa change means about 8.3 m, a 10 hPa error in the reference shifts the computed altitude by about 83 m, and on extreme days the error reaches ±100–250 m. That is why the app asks students to use the day's QNH — the one entered in **Settings**, if you have set it — instead of 1013.25 hPa when they compare with their town.

What this means for the lesson. On a high-pressure day, a group can get "we are at −40 m". The sensor is not broken and the students have not made an arithmetic mistake: the formula is behaving exactly as it should. It is also why aircraft altimeters are reset to the local pressure before every landing.

The rule to use for the rest of the activity. Absolute altitude is not the truth here; what is measured are the *differences* between floors. The difference between the ground floor and the third floor does not depend at all on the assumed reference, because the same reference pressure cancels out. That is why the graph is plotted against height *above the starting floor*, not altitude above sea level, and why you never compare one box's absolute value with another's — two BME280 units can differ by 1–2 hPa in absolute terms while still tracking the same changes flawlessly.

The right external check is not altitude but **QNH**: if a group's computed value matches the national forecast to within two or three hPa, their sensor is well calibrated.

4. **The dew point (10 min on the short route; 8 min on the full route).** Students compute T_d by hand with the Magnus formula and check it against the app's **Dew point** card — this is the best agreement they will see all lesson. Then ask what would happen to T_d, and separately to RH, if the room were heated by five degrees. Breathing gently *near*, not directly onto, the sensor produces a visible rise in humidity and is a legitimate demonstration.

5. **Heat Index (6 min — *full route only*).** Students compute HI by hand from the coefficients and compare it with the number on the card. Read the technical note below before this step: in the classroom, at 20–24 °C, the right question is not "what is the Heat Index?" but "am I allowed to use the model here?".

Technical note — right at the "Thermal comfort index (Heat Index)" step

The Heat Index is defined only for warm, humid air. The Rothfusz regression was fitted to tables valid above roughly **27 °C** and above **40**% relative humidity. **Below roughly 27 °C the value is not quoted as a result.** In an ordinarily heated classroom, at 20–24 °C, you are always outside that range.

The app still shows a number there. The formula on screen now also states its range of validity, but the **Heat Index** card is never left blank: whenever the temperature falls below 27 °C **or** the relative humidity falls below 40% — either one is enough, so 28 °C with 35% also lands here — the app shows in it *the air temperature itself*, because the regression is not defined there. That number is not a computed apparent temperature — it is T repeated. A student who copies it into the table as "Heat Index" has actually written the temperature twice. Tell them this before they write, not after.

What to do instead. On the short route, skip this step entirely. On the full route, have them compute HI by hand from the coefficients for the room's conditions, then compare the two numbers and answer a single question in writing: *is the model being used inside its valid range?* A model with an honestly stated domain of validity is better science than a formula applied everywhere — and that is, in fact, the point of this step.

6. **QNH and the calibration check (6 min — *full route only*).** Students reduce the measured pressure to sea level and compare the result with the forecast value on the board. This is the lesson's only external check: a match to within two or three hPa means a well-calibrated sensor.

7. **Comparison between two levels (12 min — *short route only*; replaces the traverse).** The box stays in the room, with the laptop beside it. Record the pressure, then carry it one floor up (or down), wait two to three minutes for the number to settle, and record again. Return to the starting floor, wait again, and record a third time. Three values, one storey, five minutes of waiting: if the first and third readings agree, the difference between them and the middle one is the signal, and it should be of order 0.3–0.4 hPa. This is where the written prediction from the start of the lesson connects to a measured number.

8. **The stairwell traverse (lesson 2, 25 min — full route and project).** One group measures the floor-to-floor height and the whole class adopts that value. Groups then carry the box and laptop to each floor in turn, wait until the pressure stops drifting, and record the floor number, cumulative height, **temperature** and pressure. Stagger the groups by one floor so that no two are ever on the same landing.

   Three readings, but only at two levels. At the ground floor and at the top floor, take **three readings**, about thirty seconds apart, *after* the pressure has settled — you do not wait another two to three minutes for each one. At the floors in between, a single reading after settling is enough. That is how it all fits into the 25 minutes of the plan: about five minutes at the ground floor, three and a half minutes at each intermediate floor, six at the top floor and seven on the way back down. The three readings at each end are the ones that give the spread to compare with the difference between levels — see the investigation box. The sheet has its rows printed exactly for this pattern.

   At the end, everyone returns to the starting floor and records again: if the value differs from the first one by more than a few hundredths of a hPa, the weather itself changed during the traverse, and the resulting gradient cannot be trusted. The return row is printed on the sheet precisely because it is not optional.

9. **Continuous monitoring (both routes; runs in the background and costs no lesson time).** Leave one box logging for the whole lesson and inspect the pressure trace at the end. Expect small changes: an ordinary tendency is 0.1–0.3 hPa per hour, that is only a few hundredths of a hPa in ten to fifteen minutes — which is why the two decimal places matter. Pressure also has a natural daily swing of up to about one hPa between maximum and minimum (maxima around 10:00 and 22:00, minima around 04:00 and 16:00), which does not mean the weather is changing. Meteorologists judge the tendency over **three hours**: below about 1 hPa in three hours it is steady, up to 3.5 hPa it is rising or falling, and above 3.5 hPa it is rapid. After at least five minutes, at the monitoring step, the app writes the **Pressure tendency** under the cards: the fitted slope in hPa/h, the change it would make over three hours and the matching word — for example *steady*. Rising pressure suggests settled weather, a sustained fall an approaching depression.

10. **Closing (8 min on the short route; 4 min on the full route).** Close with the app's conclusion step, which quotes the prediction written at the start and whose green button reads **Finish**, and ask every student for the conclusion sentence on the sheet, set against their own

prediction.

11. **Analysis and the connection outward (lesson 3 — project).** Fit the $\ln P$ against height line, extract H, and close by scaling up: the same equation explains why aircraft fly at 11 km, where the pressure is about a quarter of its sea-level value, why climbers need supplementary oxygen, and how radiosondes work.

The investigation, in one table

The question. How does air pressure change with height, and how fast?
The hypothesis. Written on the sheet **before** any measurement, with a number and a unit for the change per storey, and not changed afterwards.
What I change. Height: the floor the box is on, expressed in metres above the starting floor.
What I measure. Pressure, in hPa, to two decimal places, exactly as the app gives it. Temperature is also recorded, in its own column on the sheet, because it explains part of the reading's drift.
What I keep the same. The same stairwell, the landing doors closed, the same box from start to finish, the same waiting time at every landing, and the same person reading the screen.
How many repeats. Three readings at the ground floor and three at the top floor, taken about thirty seconds apart *after* the number has stopped drifting — you do not wait again for each one. At the intermediate floors, a single reading after settling: those give the shape of the line, not the spread. Plus the mandatory return to the ground floor at the end. The sheet has rows printed for exactly this pattern, and step 8 shows how it fits into the 25 minutes.
What makes a result valid — and what makes it an artefact. Compare the spread (maximum minus minimum) of the three ground-floor readings, and of the three top-floor readings, with the pressure difference between the two levels. **If the spread at one level is larger than the difference between two levels, the correct conclusion is that the difference is not detectable with your setup — not that it does not exist.** These are two different statements, and the distinction between them is exactly what this activity teaches. The remedy is not to hunt for better numbers in the table, but to increase the signal (more floors) or reduce the noise (a longer wait). Likewise, if the return reading does not match the starting one, the weather changed during the measurement and the gradient is affected: apply a linear correction in time, or repeat the traverse.

6. Analysis after the activity

6.1 What the data should look like

Quantity	Typical value or range	Interpretation
Indoor temperature	19 to 25 °C	Rises slowly through the lesson as the class occupies the room.
Relative humidity	30 to 60%	Lower in winter with heating on; rises noticeably if students crowd the sensor.
Station pressure	960 to 1030 hPa	Depends on both altitude and the synoptic situation.
Computed altitude	with 1013.25 hPa: true value ±100–250 m; with the day's QNH: a few metres	The error comes mainly from the assumed reference, not from the sensor.
QNH	within 2–3 hPa of the forecast	A larger gap means either a mis-measured height or a poorly calibrated sensor.
Dew point	5 to 14 °C indoors	Deficit $T - T_d$ is usually 8–15 °C.
Pressure change per storey	0.30 to 0.40 hPa	For a floor-to-floor height of 2.5–3.3 m.
Fitted scale height H	6 to 11 km	The theoretical value is 8.4 km; a result inside this band is a good outcome for an experiment done on stairs.
Pressure drift over one hour	usually 0.1 to 0.3 hPa	Above about 1 hPa per hour the weather is genuinely changing; the natural daily swing adds about one hPa over a few hours.

6.2 The graphical analysis, step by step

1. Tabulate, for each floor, the cumulative height z above the starting floor, in metres, and the pressure P, in hPa.
2. Fill in the $\ln P$ column, with P in hPa, to **five decimal places**. The whole variation across a stairwell is only about 0.0015 in $\ln P$; students who round to two decimals get a flat line and wrongly conclude that nothing happened. The column is already printed on the sheet, with a worked example above it.
3. Plot $\ln P$ (vertical) against z (horizontal), scaling the vertical axis to *only* the range of the data — an axis starting at zero destroys the effect. The axes are prepared on the sheet for exactly this reason. Draw the line of best fit and compute the gradient $m = \Delta(\ln P)/\Delta z$, in m^{-1}: a small negative number, of order -1.2×10^{-4}.
4. The scale height is $H = -1/m$. Compare it with the theoretical value $RT/(Mg) = 8.4$ km and express the discrepancy as a percentage.
5. **Extension:** rearranging gives $M = RT/(gH)$, so students can *measure the mean molar mass of air* — expect 0.02 to 0.04 $\mathrm{kg\,mol^{-1}}$ against the true value of 0.0290 $\mathrm{kg\,mol^{-1}}$. The fact that a staircase yields the molar mass of the atmosphere usually gets a reaction.

6.3 Questions to ask afterwards

1. Your computed altitude differed from the school's true altitude. Name two distinct causes and say which one is larger, with a number.

2. Why does taking the logarithm turn the pressure–height relationship into a straight line, and what would the graph of P against z have looked like over the same range?
3. The scale height is about 8.4 km, yet aircraft fly at 11 km and Everest is nearly 8.8 km high. What fraction of sea-level pressure remains at each of those heights, and what does that imply for a climber?
4. Relative humidity fell during the lesson while the dew point stayed nearly constant. Explain how both can be true at the same time.
5. Two schools 30 km apart report station pressures of 998 and 1006 hPa. Can you conclude the weather differs between them? What must you calculate before answering?

6.4 Assessment ideas along the way

- **Low stakes:** the student completes a partially filled table of P and $\ln P$ for a supplied data set and states, in one sentence, what the gradient represents.
- **Standard:** a two-page report — prediction, method, the $\ln P$ against z graph with the gradient drawn, the value of H with its percentage discrepancy from theory, and one named source of systematic error with an estimate of its size.
- **Extension:** an independent derivation of the barometric formula from the hydrostatic equation, then using the fitted H to estimate the mean molar mass of air and discussing which way the answer is biased; or a least-squares fit with a computed R^2 instead of a ruler-drawn line; or logging pressure at home for a week and correlating the trend with published forecasts.

Common pitfalls and misconceptions

"The sensor is broken — it says we are at −40 m." The barometric formula assumes a sea-level pressure of exactly 1013.25 hPa. On a high-pressure day, the computed altitude genuinely comes out negative, and the formula is behaving correctly. This is the most common reaction, and it is a teaching opportunity, not a fault.

Rounding that erases the whole effect. Across a stairwell, $\ln P$ changes in the fourth decimal place. Students must carry five decimals through the whole calculation. Enforce this before they start, not afterwards.

Confusing the two lapse rates. The *dry adiabatic lapse rate* (9.8 °C/km) applies to a parcel of air that is being lifted; the *environmental lapse rate* ($\approx$ 6.5 °C/km) describes the ambient profile actually measured. The barometric formula uses the second.

Reading the sensor before it has settled, or comparing boxes. A box carried into a cold stairwell reads a drifting temperature for a minute or more. Separately, two BME280 units can differ by 1–2 hPa in absolute value while still tracking the same changes flawlessly: compare *differences*, never absolute values, between boxes.

Believing relative humidity measures the amount of water. It measures the fraction of saturation, which depends on temperature. Heating the room lowers RH without removing a single water molecule from the air — the dew point proves it.

Quoting the Heat Index value below about 27 °C. The regression is not valid there, and the app shows the air temperature itself in the card. See the technical note beside that step.

Weather changing during the traverse. A passing front can move the pressure more than the whole stairwell effect. The return-to-start check detects this; if the two readings do not match, apply a linear correction in time or repeat the measurement.

Safety and classroom management

The classroom part of this activity carries low risk. The stairwell traverse does not, and must be managed.

Nobody reads a screen while walking on stairs. The box is **carried**; it is read **standing still**. State the rule plainly before leaving the room, and repeat it at the first slip: climb, stop on the landing, only then read the screen. One named student carries the equipment, with both hands on it, and nobody runs. Groups keep to the wall side of the staircase and never stop in a doorway or on the steps themselves — only on landings. Check that the traverse does not fall during a break or a lesson changeover.

Fire doors, access and cables. Agree the route with the building manager in advance and do not prop fire doors open; roof access is not part of the activity. A USB cable between box and laptop is a trip hazard on stairs: use a short one, or tape it down.

Supervision. Groups do not move through the building unsupervised. Either stagger them so you can see the whole route, or run the traverse as a single class movement, with every group recording at the same stops.

Equipment. The sensor must never get wet; condensation inside the humidity element gives false readings for hours afterwards.

Adaptations for pupils with special educational needs

The $\ln P$ column and the axes, already prepared. At this level, arithmetic is not what is being taught; the gradient is. The student sheet comes with the $\ln P$ column printed, a worked example above the table and a grid of axes already drawn, so the student does not lose the lesson to drawing axes. For students who trip over the calculation, go one step further: **print the sheet with the $\ln P$ values already filled in** (taken from a trial run of your own, or from the exported `.xlsx` file, with a computed column), so that the student's task is only: plot the points, draw the line, read the gradient. They reach the same scientific result — H — by a different route.

The settled route, for students with motor difficulties. The whole building does not need to be traversed for there to be a result. **Two levels are enough**: the room's pressure, the pressure one floor up, and the return. That is three readings, a single round trip and the same physical conclusion, only with a smaller signal. The student can stay in the room, and a classmate carries the box on their instruction.

The role of data monitor. A student who cannot walk the stairwell is not a student who does not take part. The **data monitor** stays in the room with the group's table and the reference laptop: receives the values from each landing by phone or through classmates, enters them into the table, checks on the spot whether the three repeats at one level are close together, and tells the group if a reading needs repeating. This is the role that decides which data are valid, so it carries the greatest scientific weight in the group — not a consolation role. Two other real roles can also be held from a seat: the *screen reader*, who says when the number has stopped drifting, and the *calculator*, who keeps the $\ln P$ column and checks the decimals.

Oral answers. The conclusion, the hypothesis and the answer to the validity-range question can be given **out loud** instead of written, with the teacher or a classmate writing them on the sheet. For the sentence frames on the sheet, a student may complete them orally and only sign the row. What is assessed is the reasoning, not the speed of writing.

Language and cognitive support. Use the same words all lesson (pressure, height, gradient, scale height) and write them on the board at the start. For students who get lost in the derivation, the short route — without derivation — is the default option, not a fallback: the measured result is the same.

If things do not go to plan

You have no access to a stairwell. Use the height difference between the floor and a tabletop, or between the floor and a cupboard: about a metre, so about 0.12 hPa. The signal is small, so extend the waiting time to four or five minutes at each position and take five repeats instead of three. If the spread stays larger than the difference, that is the lesson's honest result — the difference is not detectable with this setup — and it is a better lesson in method than a neat number obtained by guessing.

You have twenty minutes, not fifty. Keep exactly three moments: the written prediction, the comparison between two levels, and the conclusion said out loud. Drop the dew point, QNH and the traverse. The written prediction at the start and the measured number at the end are all that is needed for the lesson to have a result.

The sensor will not settle. Stop and make that the topic: why does the pressure reading not jump straight to the new value? Not because the chip warms up — the pressure is temperature-compensated inside the sensor — but because the reading is *smoothed*: the sensor's filter and the app's averaging trade speed for lower noise, and any smoothing means a delay. In a stairwell, draughts add their own noise of a few hundredths of a hPa. Wait another two minutes, with the box out of anyone's hand and set on a surface, away from a radiator or sunlight. If it still will not settle, a draught in the stairwell is the cause: close the landing doors.

You have only one box. Do the traverse once, with the whole class, as a single movement. At each landing a different student reads the screen out loud and another confirms that the number has stopped moving; everyone records in their own table. You get a single data set, but the whole class analyses it, and the reading role rotates.

The box will not connect. Do not stop the lesson. Use the "What the data should look like" table in this guide as a printed data set: students write their predictions first, then receive the values row by row and work through the graphical analysis on them. Repeat the measurement on another day, after checking the cable, the port selected in the app and, if you are on WiFi, the network.

7. Further resources

Stable, well-established starting points. Search within each site for the topic named — these are root addresses, not deep links, so they will not rot over time.

- **National Meteorological Administration (ANM)** — `https://www.meteoromania.ro` — the QNH value and the synoptic chart for the day of the lesson; this is the external reference against which the class compares its result, and the only calibration check available in the room.
- **NOAA** — `https://www.noaa.gov` — search for "standard atmosphere", "altimeter setting" and "heat index". The authoritative source for the Rothfusz regression and for the definition of the reduction to sea level used in the QNH step.
- **PhET Interactive Simulations** — `https://phet.colorado.edu` — search for "gas properties" and "states of matter". The gas simulation makes the link between molecular motion, pressure and the equation of state concrete, before the derivation.
- **NASA** — `https://www.nasa.gov` — search for "Earth atmosphere model" and "atmospheric layers". Gives the standard-atmosphere equations in the same algebraic form used here, useful for students checking their own derivation.
- **Institute of Physics** — `https://www.iop.org` — search for "fluid pressure" and "practical physics". Teacher-facing guidance on hydrostatics practicals and on working with exponential relationships at upper-secondary level.

- **STEM Learning** — `https://www.stem.org.uk` — search for "logarithmic graphs" and "weather". Ready-made resources on linearising exponential data, the transferable skill this activity actually teaches.

8. Assessment

The exit ticket, identical to the one on the student sheet: *one thing that surprised you today, and one thing you would do differently if you repeated the measurement.* Collected in the last two minutes of the lesson, and it says more about the class's understanding than any mark.

For a more formal judgement, use the **shared three-strand rubric** — *measurement practice*, *data interpretation* and *scientific reasoning*. At Scientist level, the three strands look like this:

- **Measurement practice** — the student waits for the reading to settle before recording, takes repeats at each level, records five decimal places where the effect requires them, and makes the return to the starting floor without being reminded.
- **Data interpretation** — the student linearises correctly by taking the logarithm, chooses a vertical scale that suits the data, extracts the gradient with its unit, and obtains H from the gradient.
- **Scientific reasoning** — the student compares the obtained H with the theoretical value, names a source of systematic error and says which way it pushes the result; and distinguishes "the difference is not detectable" from "the difference does not exist".

The rubric itself, with descriptors for all three levels, is defined once in the *curriculum plan* and is not repeated in the guides.

9. The student sheet

The sheet the students fill in is a separate document: `docs/en/student_sheets/physics_of_the_atmosphere.pdf`. It has two pages — the prediction, the baseline measurement and the traverse table on the first; the return check, the graph, the gradient and the conclusion on the second — so it prints double-sided, one sheet per student. It is built on the three layers — what I see on the screen, what I do on my own, the conclusion I write — asks for the prediction **before** any measurement, and comes with the table already labelled, with the $\ln P$ **column printed**, a **temperature column** and the **axes already drawn**. The "ground floor, return" row is printed as a mandatory row, not an optional one, and below the table there is a separate block for the **three readings at the ground floor and at the top floor**, with the spread to compare against the difference between levels. The table is not repeated here precisely so the two cannot drift apart: there is a single printable version, and it is the sheet.